Introduction to Artificial Intelligence and Machine Learning (CSN11601)

LAB TASK-4

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Problem Statement 1: Student Text Analysis
Scenario:
You are working as a data analyst in your university library. The library provides you with a text file
containing the content of several research articles. Your task is to write a Python program that:
1. Reads the provided text file.
2. Prints all unique words in the file in alphabetical order.
3. Allows the user to write a summary of the file content into another text file.
Sample Text File (research_articles.txt):
Python is a popular programming language.
It is widely used in data science, web development, and automation.
Python programming helps in data analysis and machine learning.
Expected Output:
['a', 'and', 'analysis', 'automation', 'data', 'development', 'helps', 'in', 'is', 'language', 'learning',
'machine', 'popular', 'programming', 'python', 'science', 'used', 'web', 'widely']
Tasks for Students:
• Implement read and write operations in Python.
• Ensure the output list of words is unique and alphabetically sorted.

A1. CODE:

```
f=open('A1.txt','r')
ab=f.read()
l1=[]
for i in ab.split():
        if i not in l1:
                l1.append(i)

l1.sort()
print("All unique words in the file in alphabetical order")
print(l1)
f.close()
summary=input("Enter the summary you want to save into another file : ")
f=open('summary.txt','w')
f.write(summary)
f.close()
```

OUTPUT:

```
(base) user@wipro-desktop:~/Aayansh_20253004/Lab_Task4$ python p1.py
All unique words in the file in alphabetical order
['It', 'Python', 'a', 'analysis', 'and', 'automation.', 'data', 'development,', 'helps', 'in', 'is', 'language.', 'learning.', 'machine', 'po
pular', 'programming', 'science,', 'used', 'web', 'widely']
Enter the summary you want to save into another file : The file is really good
```

Files:

A1.txt

```
1 Python is a popular programming language.
2 It is widely used in data science, web development, and automation.
3 Python programming helps in data analysis and machine learning.
```

Summary.txt

```
1 The file is really good
```

Problem Statement 2: Multi Format data processing
Scenario:

You are an intern at a data management company. You are given multiple files containing sales and
customer data in different formats (CSV, TSV, Excel, Text). Your task is to:
1. Read the data from all the given files.
2. Copy the content of each file into a new file (e.g., output.csv, output.txt, etc.).
3. Print the first 5 lines of each file to verify its content.

A2. CODE:

```
import pandas as pd

a2=pd.read_csv('A2.csv')
a3=pd.read_csv('A3.tsv',sep='\t')
a4=pd.read_excel('A4.xlsx')
a5=open("A5.txt",'r').read()

a2.to_csv("output.csv",index=False)
a3.to_csv("output.tsv",sep='\t',index=False)
a4.to_excel("output.xlsx",index=False)
open("output.txt",'w').write(a5)

print("\nA2.csv (first 5 rows):\n",a2.head())
print("\nA3.tsv (first 5 rows):\n",a3.head())
print("\nA4.xls (first 5 rows):\n",a4.head())
print("\nA5.txt (content):\n",a5[:500])
```

OUTPUT:

```
(base) user@wipro-desktop:~/Aayansh_20253004/Lab_Task4$ python p2.py

A2.csv (first 5 rows):
    ID   Name   Product   Price
0   1   John  Keyboard   29.99
1   2   Anna     Mouse   19.99
2   3  David   Monitor  159.99

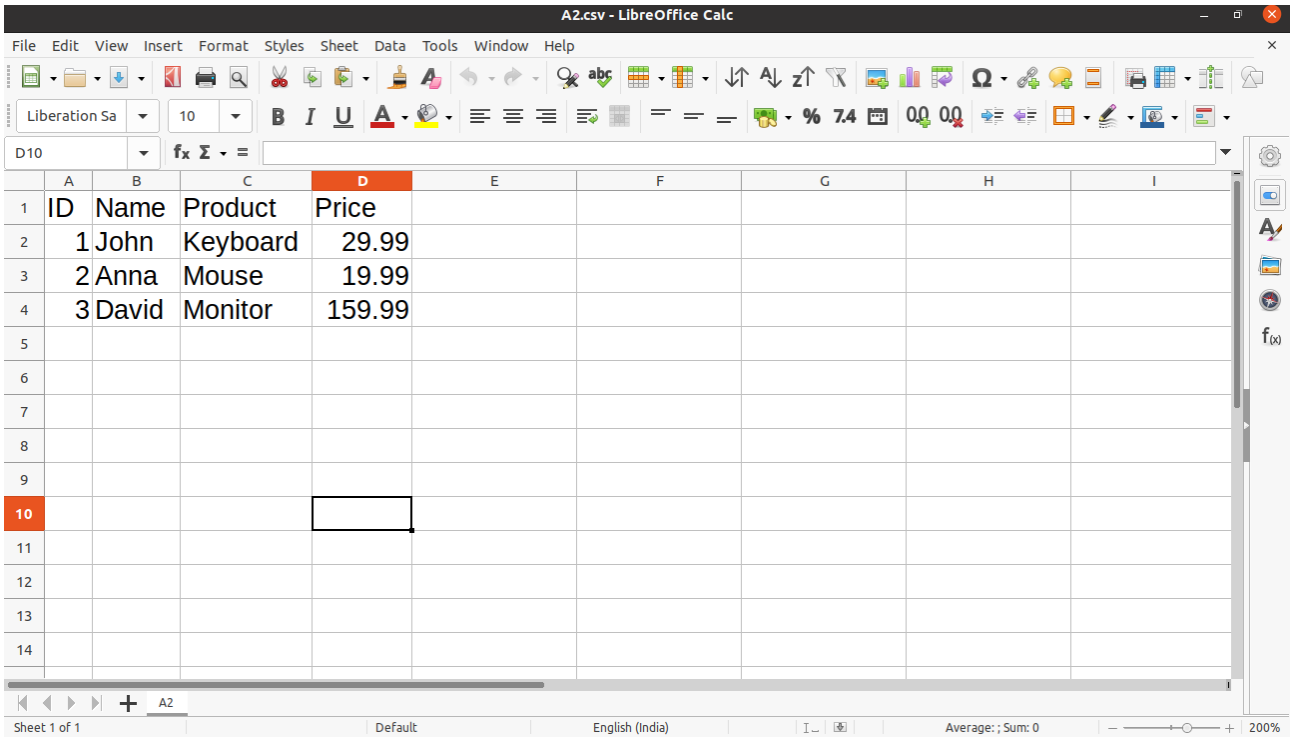
A3.tsv (first 5 rows):
    ID   Name              Email
0   1   John   john@example.com
1   2   Anna   anna@example.com
2   3  David  david@example.com

A4.xls (first 5 rows):
    ID   Product  Quantity   Price
0   1  Keyboard        50   29.99
1   2     Mouse       100   19.99
2   3   Monitor        20  159.99

A5.txt (content):
 Meeting notes:
- Discussed project timelines
- Assigned tasks to team members
- Next meeting on Monday
```

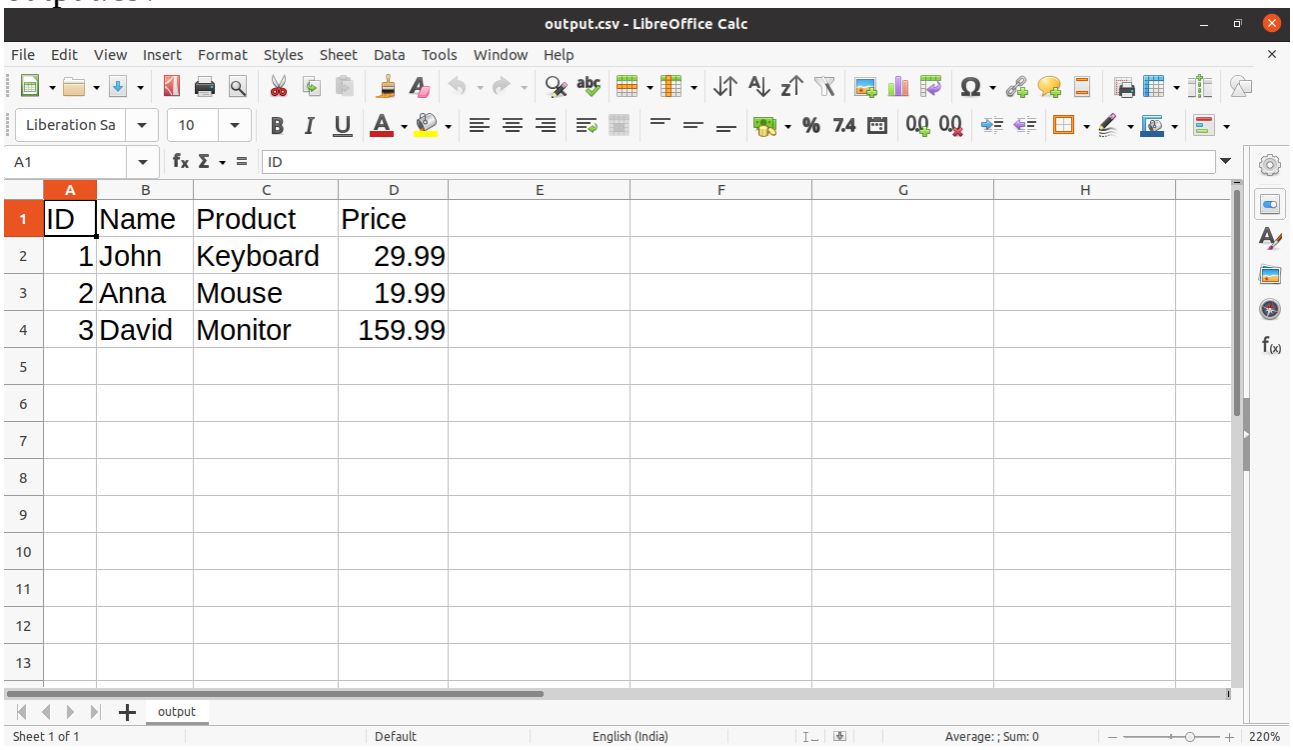
Files:

A2.csv



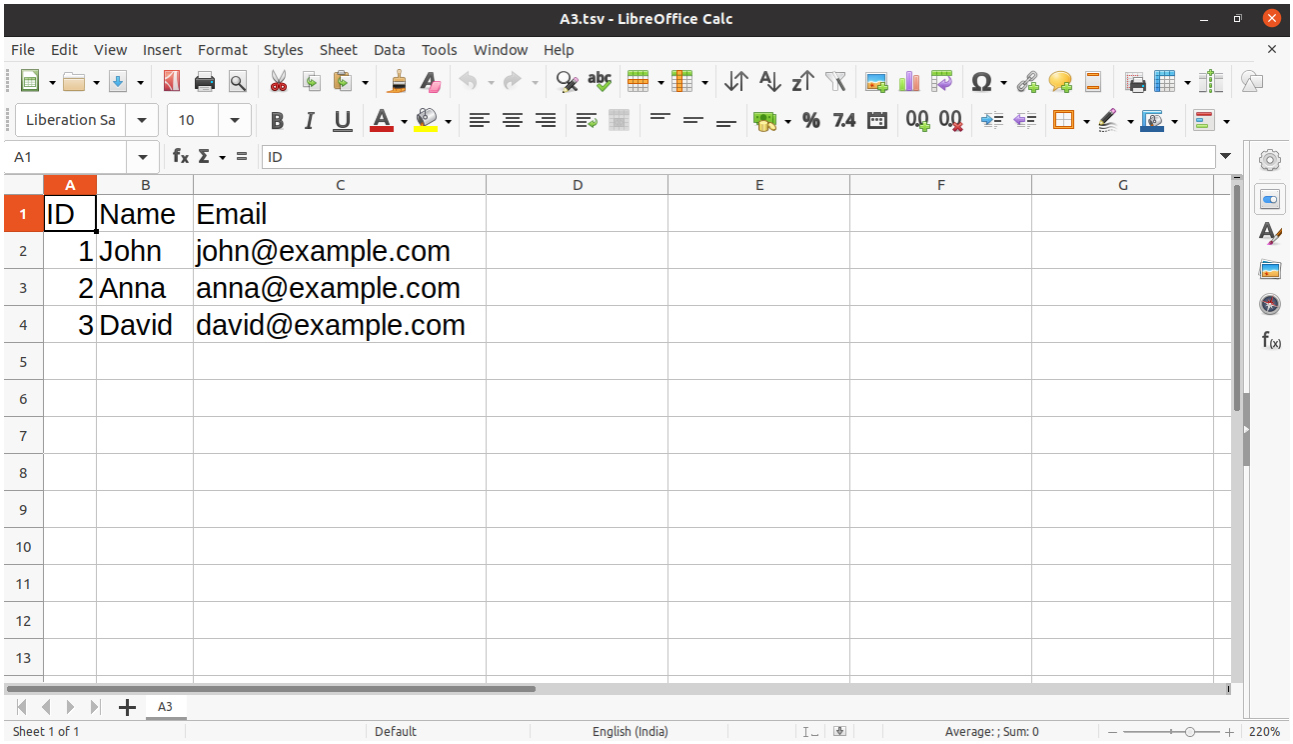
	A	B	C	D
1	ID	Name	Product	Price
2	1	John	Keyboard	29.99
3	2	Anna	Mouse	19.99
4	3	David	Monitor	159.99

output.csv



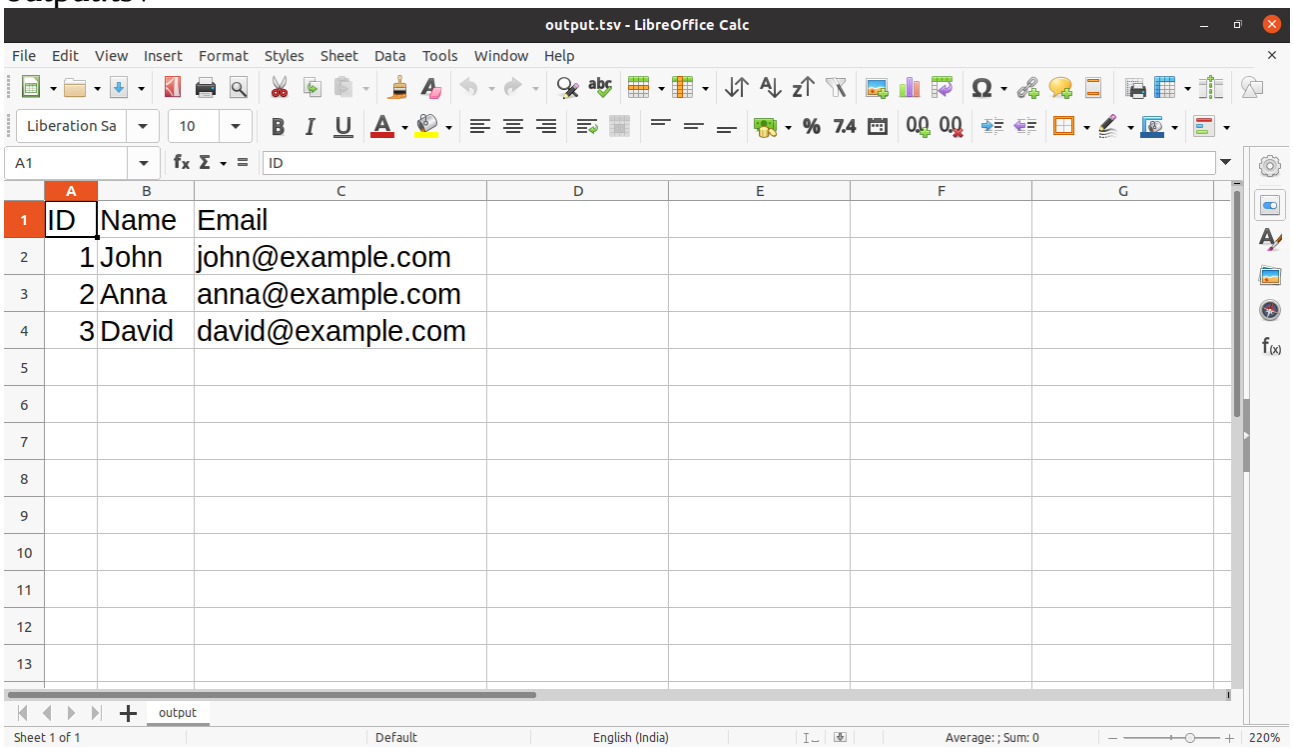
	A	B	C	D
1	ID	Name	Product	Price
2	1	John	Keyboard	29.99
3	2	Anna	Mouse	19.99
4	3	David	Monitor	159.99

A3.tsv



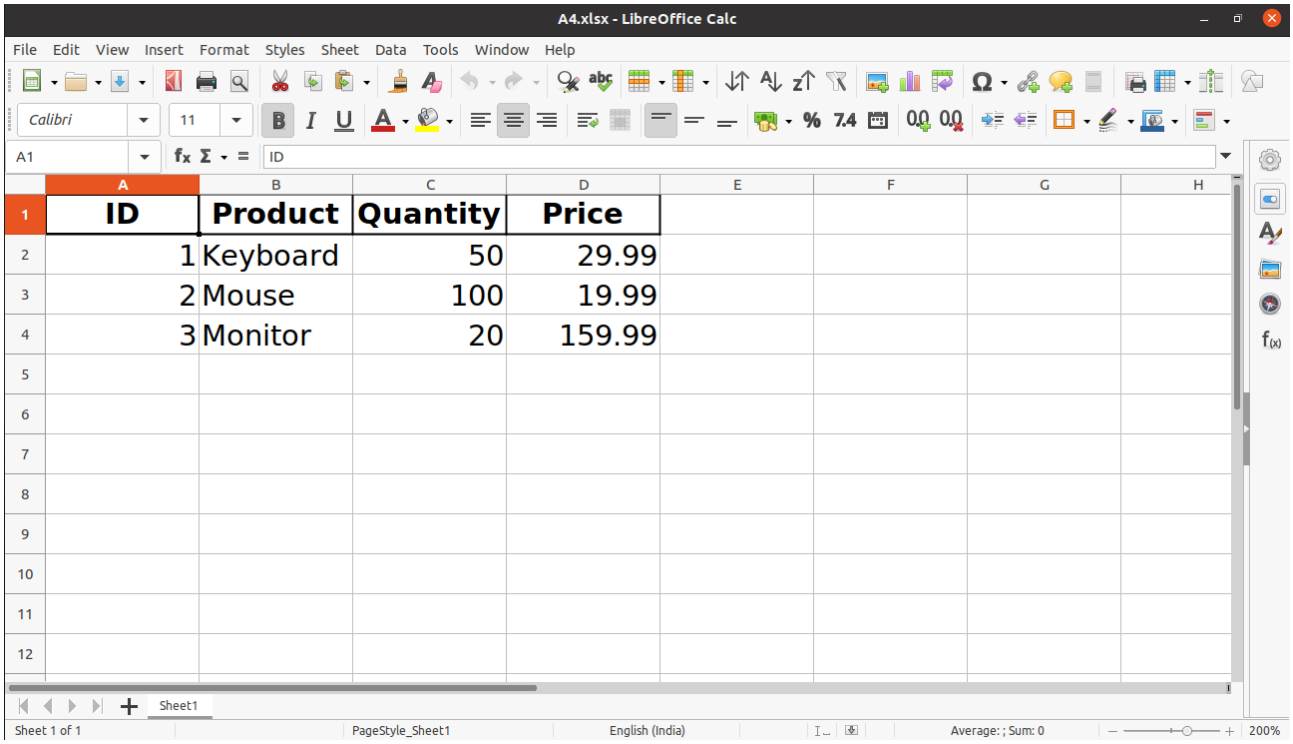
ID	Name	Email
1	John	john@example.com
2	Anna	anna@example.com
3	David	david@example.com

output.tsv



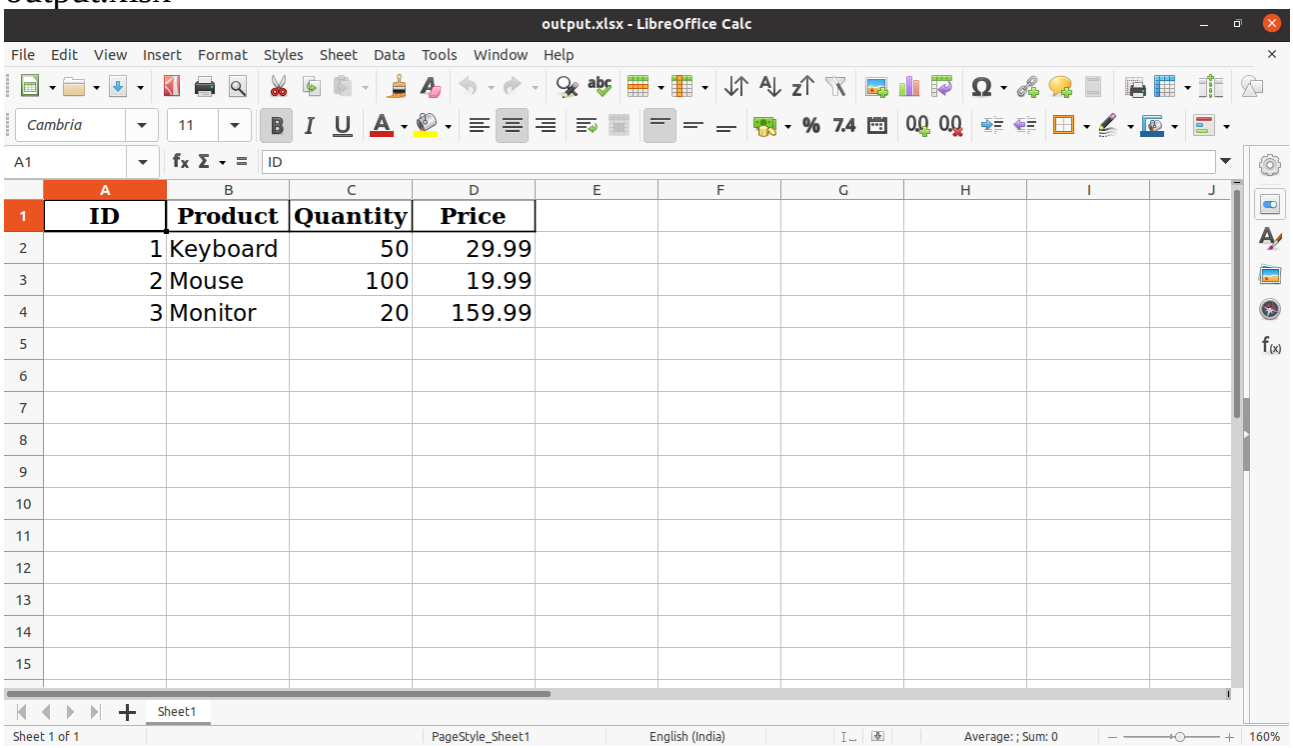
ID	Name	Email
1	John	john@example.com
2	Anna	anna@example.com
3	David	david@example.com

A4.xlsx



ID	Product	Quantity	Price
1	Keyboard	50	29.99
2	Mouse	100	19.99
3	Monitor	20	159.99

output.xlsx



ID	Product	Quantity	Price
1	Keyboard	50	29.99
2	Mouse	100	19.99
3	Monitor	20	159.99

A5.txt

```
Meeting notes:
- Discussed project timelines
- Assigned tasks to team members
- Next meeting on Monday
```

output.txt

```
Meeting notes:
- Discussed project timelines
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```